Homework 07

Steven Jia

October 11, 2026

Question 1: LDA

(a) Estimate the LDA components

There are 462 observations, 9 numerical predictors, and 2 classes. Family history is coded as 1 for Absent and 2 for Present. The empirical priors and pooled covariance estimate are

$$\hat{\pi}_k = \frac{n_k}{n}, \qquad \widehat{\Sigma} = \frac{1}{n-K} \sum_{k=0}^{1} \sum_{i:y_i=k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^\mathsf{T}, \qquad n - K = 460.$$

Table 1: Empirical class priors.

Class	Count	Prior
0	302	0.65368
1	160	0.34632

Table 2: Class-specific predictor means.

Predictor	Class 0	Class 1
sbp	135.4603	143.7375
tobacco	2.6347	5.5249
ldl	4.3442	5.4879
adiposity	23.9691	28.1202
famhist	1.3179	1.6000
typea	52.3675	54.4938
obesity	25.7375	26.6229
alcohol	15.9314	19.1452
age	38.8543	50.2938

The following table contains the complete pooled covariance matrix. The abbreviations tob, adi, fh, typ, obe, and alc denote tobacco, adiposity, famhist, typea, obesity, and alcohol, respectively.

Table 3: Pooled covariance matrix (denominator 460).

	sbp	tob	ldl	adi	fh	typ	obe	alc	age
sbp	405.435	14.585	4.581	49.164	0.337	-15.588	18.939	64.400	95.134
tob	14.585	19.243	0.763	7.538	0.016	-2.057	1.833	20.517	22.765
ldl	4.581	0.763	4.001	6.033	0.092	0.345	2.660	-2.533	6.479
adi	49.164	7.538	6.033	56.753	0.433	-5.310	22.708	16.119	60.509
fh	0.337	0.016	0.092	0.433	0.226	0.081	0.184	0.768	0.997
typ	-15.588	-2.057	0.345	-5.310	0.081	95.566	2.640	7.960	-20.278
obe	18.939	1.833	2.660	22.708	0.184	2.640	17.615	4.689	15.697
alc	64.400	20.517	-2.533	16.119	0.768	7.960	4.689	598.277	27.886
age	95.134	22.765	6.479	60.509	0.997	-20.278	15.697	27.886	184.132

(b) Calculate LDA scores and in-sample fitted labels

For each observation, the fitted label is the class with the largest computed score:

$$\hat{\delta}_k(\mathbf{x}_i) = \mathbf{x}_i^\top \widehat{\Sigma}^{-1} \hat{\mu}_k - \tfrac{1}{2} \hat{\mu}_k^\top \widehat{\Sigma}^{-1} \hat{\mu}_k + \log \hat{\pi}_k.$$

Table 4: Dimension checks for the manual calculation.

Object	Dimensions
X	462 x 9
y	length 462
Class means	2 x 9
Pooled covariance	9 x 9
B	9 x 2
Intercepts / priors	length 2
Scores	462 x 2
Fitted labels	length 462

Table 5: Manual LDA: observed classes in rows, predicted classes in columns.

	0	1
0	258	44
1	73	87

The in-sample misclassification rate is $117/462 = 25.32\%$.

(c) Compare with `MASS::lda`

Table 6: MASS::lda: observed classes in rows, predicted classes in columns.

	0	1
0	258	44
1	73	87

Matching fitted labels: 462 out of 462; empirical priors and class means also agree.

The results of my manual LDA calculation and `MASS::lda` were exactly the same. All 462 observations had the same predictions, and the two confusion matrices were also identical. This is because both methods used the same data and variables, as well as the same class priors and pooled covariance estimate.

Question 2: QDA and marginal screening

(a) QDA using the 40 largest training marginal variances

The training and test sets contain 7291 and 2007 observations, respectively, with all ten digit classes. The 40 predictors are selected by their marginal sample variances over the training observations. The same pixel indices and original pixel values are used in the test set. Class priors are empirical training proportions.

Table 7: QDA with 40 pixels: observed rows and predicted columns.

	0	1	2	3	4	5	6	7	8	9
0	319	0	24	6	2	2	1	0	5	0
1	0	238	5	0	11	0	7	0	3	0
2	8	0	168	8	8	3	0	0	3	0
3	5	0	4	128	0	14	0	0	15	0
4	0	2	13	1	150	1	0	1	7	25
5	12	0	3	5	1	131	0	0	6	2
6	0	0	8	0	1	8	148	0	5	0
7	1	0	6	1	6	0	0	124	2	7
8	1	0	10	12	3	2	0	0	137	1
9	0	0	1	2	7	1	1	4	9	152

Test misclassification rate (40 predictors): **312/2007 = 15.55%**.

(b) Check whether the method works

The ranks below are computed from centered within-class predictor matrices. A full rank of p means the corresponding $p \times p$ sample covariance is nonsingular. The QR rank calculation uses the same default tolerance as ordinary `MASS::qda`.

Table 8: Within-class rank checks before fitting QDA.

Digit	Training n	Rank (256)	Rank (40)	Rank (10)
0	1194	253	40	10
1	1005	126	40	10
2	731	254	40	10
3	658	256	40	10
4	652	251	40	10
5	556	256	40	10
6	664	237	40	10
7	645	224	40	10
8	542	251	40	10
9	644	225	40	10

I found that QDA could run successfully after selecting 40 pixels, with a test error rate of 15.55%. When all 256 pixels were used, the covariance matrices for some classes could not be inverted, which prevented QDA from completing the calculation. After selecting 40 pixels, the covariance matrix for every class could be inverted.

(c) Repeat with the 10 largest-variance predictors

The 10 selected pixels are the first 10 in the same ranking used in part (a), so this predictor set is a subset of the 40-pixel set.

Table 9: QDA with 10 pixels: observed rows and predicted columns.

	0	1	2	3	4	5	6	7	8	9
0	298	0	11	1	1	16	28	4	0	0
1	0	241	10	0	3	1	1	0	2	6
2	19	2	95	14	14	3	22	21	5	3
3	5	0	6	103	3	42	0	0	7	0
4	1	5	14	0	95	2	4	15	17	47
5	15	0	6	25	4	96	4	2	8	0
6	41	0	15	4	0	18	78	7	5	2
7	4	1	6	0	6	0	1	122	0	7
8	1	2	11	36	12	11	4	1	74	14
9	0	2	9	2	20	0	0	16	10	118

Test misclassification rate (10 predictors): **687/2007 = 34.23%**. The 10-pixel error exceeds the 40-pixel error by 18.68 percentage points.

I found that when the number of variables decreased from 40 to 10, the test error rate increased from 15.55% to 34.23%. Although fewer variables meant that the model needed to estimate fewer parameters, some information for recognizing the digits was also lost. Using only 10 pixels may not be enough to distinguish these digits. Therefore, in this experiment, I would prefer to use 40 variables because the 40-variable model had a lower test error rate.

Question 3: Compare LDA and QDA

(a) One experiment at each value of ρ

Seed: **432073**. RNG: **Mersenne-Twister / Inversion / Rejection**. Training size: **100 per class**; independent test size: **1,000 per class**. Equal class priors: **0.5, 0.5**. All **10 predictors** enter both classifiers.

Table 10: One independently generated dataset at each rho.

rho	LDA test error	QDA test error
0.0	0.1670	0.1975
0.2	0.1690	0.2020
0.4	0.1660	0.1975
0.6	0.1730	0.1745
0.8	0.1600	0.1340
0.9	0.1775	0.1020

(b) Fifty experiments at each value of ρ

Table 11: Mean error across 50 new experiments per rho; MCSE = SD of experiment errors / sqrt(50).

rho	LDA mean error	QDA mean error	LDA MCSE	QDA MCSE
0.0	0.1682	0.1960	0.0014	0.0020
0.2	0.1721	0.1991	0.0013	0.0018
0.4	0.1717	0.1939	0.0016	0.0018
0.6	0.1716	0.1789	0.0016	0.0014
0.8	0.1670	0.1366	0.0012	0.0013
0.9	0.1685	0.1002	0.0014	0.0011

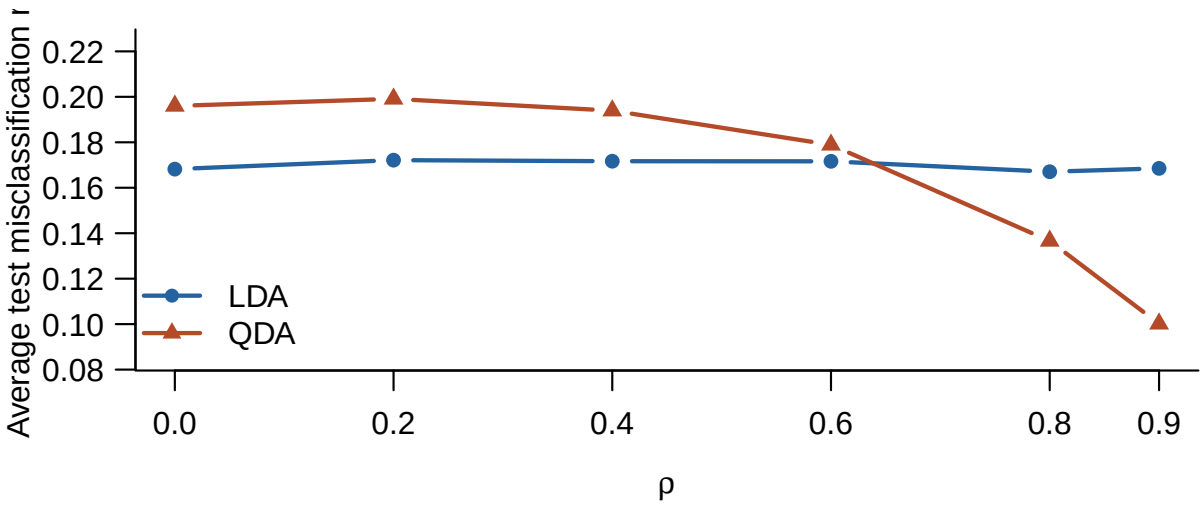


Figure 1: Average test error across 50 independent experiments at each rho.

(c) Interpretation

I found that when ρ was small, LDA had a lower average test error rate than QDA. For example, when $\rho = 0$, the error rate was 16.82% for LDA and 19.60% for QDA. In these settings, the covariance matrices of the two classes were the same or similar. LDA used a shared covariance matrix and needed to estimate fewer parameters, so its results were more stable.

When ρ was large, QDA performed better. The first two variables were positively correlated in the first class and negatively correlated in the second class, so the covariance matrices of the two classes differed more. QDA could estimate them separately, so it was more suitable.

This reflects the bias-variance trade-off. LDA's results were more stable, with lower variance, but its bias could be larger when the covariance matrices of the two classes differed greatly. QDA could estimate the covariance matrices separately and could have lower bias, but its results were also more sensitive to changes in the training sample.

Although the other eight variables could not help distinguish the classes, the model still needed to estimate their means and covariances. These estimates were affected by random variation in the sample, which could make predictions worse. For example, QDA needed to estimate more parameters, so these variables could affect it more.

Question 4(a), (c): Skill definition, invocation and review

Skill: `lda-qda-report`. **File size:** 4130 bytes (limit: 5,000 bytes). The complete, dataset-independent skill appears after the analysis report. It is kept in this homework folder, outside automatically loaded skill directories, and was supplied through an explicit invocation.

Recorded invocation prompt:

> Use $lda-qda-report from lda-qda-report/SKILL.md to analyze data/saheart.csv in R. The response is chd; use all nine other variables as numerical predictors, coding famhist as 1 for Absent and 2 for Present. Use seed 432074. Seek the lowest cross-validation misclassification rate within the skill's fixed LDA/QDA candidate grid, including ordinary LDA and QDA. Reserve the test set before model selection, use the same training folds for all candidates, and fit preprocessing within each fold. Refit the selected model on all training observations and evaluate it once on the reserved test set. Produce a one-page report draft with a figure, a table and numerical fitting results. Leave explanatory English prose as labeled placeholders and supply Chinese references separately.

Review. I checked whether the different models used the same cross-validation folds and whether standardization and PCA used only the corresponding training data. I found that these steps met the requirements, so I did not modify the skill this time. I selected the model using only training and cross-validation results and did not use the test results to select a different model.

The following page contains the one-page analysis report. Its one-page limit excludes this invocation/review section and the complete skill appendix.

Question 4(b): One-page SAheart analysis report

Data and procedure. SAheart has 462 complete observations and nine numerical predictors; `chd` is 0/1 and `famhist` is Absent=1, Present=2. No observations were excluded. Seed 432074 produced a stratified training/test split of 323/139. The fixed grid comprises ordinary LDA/QDA, standardized PCA with 1–8 components followed by either method, and three probability blends (LDA weights .25/.50/.75). Five repeats of stratified five-fold CV used identical folds, with preprocessing and empirical priors fitted within each fold. All 21 candidates were valid. Training class counts (0/1) are 211/112; full and fold covariance ranks are 9.

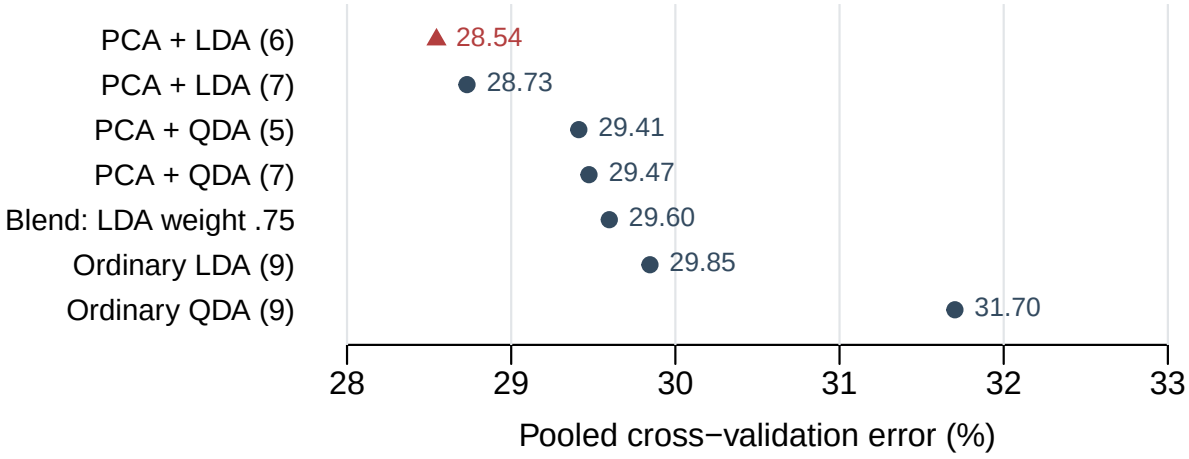


Figure 2: Five lowest CV errors and ordinary comparisons. Full results for all 21 candidates are in the companion CSV.

Model	Dimension	CV errors	CV error (%)
Ordinary LDA	9	482/1615	29.85
Ordinary QDA	9	512/1615	31.70
Selected: PCA + LDA (6)	6	461/1615	28.54

Final fit and held-out evaluation. The selected standardized PCA + LDA model retains six dimensions (85.37% of standardized predictor variance), with fitted class priors (0/1) 0.6533/0.3467. Its once-evaluated test error is **30/139 = 21.58%**, with 95% Wilson interval [**15.56%, 29.14%**]. The confusion matrix uses observed rows and predicted columns:

	0	1
0	82	9
1	21	27

Findings. In this comparison, I chose LDA with six PCA components because it had the lowest cross-validation error rate, at 28.54%. Ordinary LDA and QDA had cross-validation error rates of 29.85% and 31.70%, respectively. Finally, I evaluated the selected model on the test set, with an error rate of 21.58%.

Limitations. I found that class 0 had more observations and class 1 was misclassified more often, so the overall error rate did not mean that both classes were predicted well. LDA used a shared covariance matrix, while QDA estimated a separate one for each class. However, the data might not satisfy these model assumptions or follow a normal distribution.

Appendix: Complete contents of `SKILL.md`

```
---
name: lda-qda-report
description: Run a reproducible LDA/QDA comparison and write a one-page statistical report only when this skill is
     explicitly requested.
---

# LDA/QDA report

Load this file only on explicit invocation. Keep it outside automatically
loaded skill directories. Receive the dataset, response, predictors and
categorical coding separately; never embed a particular dataset or its results
in these instructions.

## Analysis

- Use R and reproducible code. Record the seed, software versions, sample sizes,
  coding, class counts and excluded observations. Check missingness, numerical
  values, class balance and covariance rank. Explain any fitting failures.
- Before model selection, reserve about 30% of observations as a stratified
  test set. Set the seed once and retain observation IDs. Do not inspect test
  performance to change candidates, settings, preprocessing or this skill.
- Include ordinary LDA (one pooled covariance) and ordinary QDA (one covariance
  per class), with empirical class priors estimated from fitting observations.
  Keep every classifier based on LDA, QDA, or their probability combination.
- Define the complete candidate grid before cross-validation: ordinary LDA and
  QDA; standardized PCA followed by either method with dimensions
  1 through min(p - 1, 10); and arithmetic probability blends of ordinary LDA
  and QDA with LDA weights 0.25, 0.50 and 0.75. Omit PCA when p = 1.
  Use the class with maximum predicted probability, with a fixed class-order
  tie rule. Do not add other classifiers or silently regularize failed models.
- Use five repeats of stratified five-fold cross-validation on training data,
  with identical folds for all candidates. If classes are too small, reduce
  the fold count and explain it. Within each fold, estimate all preprocessing
  and priors only from its fitting observations. PCA requires fitting-fold
  centering, sample-SD scaling and loadings; transform validation data using
  those values. Fit both blend components on the same fitting observations.
- Compare misclassification rates pooled across validation predictions.
  Declare a candidate invalid if a fold cannot be fitted; retain its failure
  reason. Select the valid candidate with the lowest CV error. For exact ties,
  prefer fewer retained dimensions, then LDA, then a blend, then QDA; resolve
  any remaining tie by the recorded candidate order. Report every candidate's
  CV result in the reproducible companion, without treating the selected CV
  minimum as an unbiased estimate of final prediction error.
- Before using the test set, save the candidate grid, folds, CV results and
  selected settings. Refit that selected classifier and its preprocessing on
  all training observations. Evaluate it once on the reserved test set;
  report error count/denominator, misclassification rate, confusion matrix
  (observed rows, predicted columns) and a 95% Wilson binomial interval.
  Any later model or skill revision must use training/CV evidence and needs a
  fresh untouched test set for another final evaluation.

## Report

Produce an analysis report of at most one printed page in the user's document
format. Use readable text and labels; shorten prose rather than shrink fonts.
Include:

1. Data/coding, seed, split counts and candidate/CV procedure.
2. A figure comparing candidate CV error rates with meaningful labels.
3. A compact table with ordinary LDA, ordinary QDA and the selected model,
   including CV errors and the selected model's fitting dimension.
4. Numerical fitting results (training class counts, fitted priors and retained
   dimension), selected settings, held-out errors and their uncertainty.
5. A brief explanation of the findings, covariance assumptions, selection
   uncertainty, class imbalance and distributional limitations.

Keep the complete skill, invocation prompt, code and detailed results outside
the one-page report. If the user wants to write the English prose, supply a
Chinese reference separately and use clearly labeled English placeholders
in the draft. Recheck the one-page limit after their prose is inserted.
```